Do Now · retrieval

Lesson 142 · Networks: LANs, WANs, Standards and Protocols

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which ACID property keeps committed transaction changes after a crash?

Q2

How many bits are in one byte?

Q3

What data structure uses LIFO?

Q4

What is a trace table used for?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Identify scope — LAN connects local devices; WAN connects distant sites.

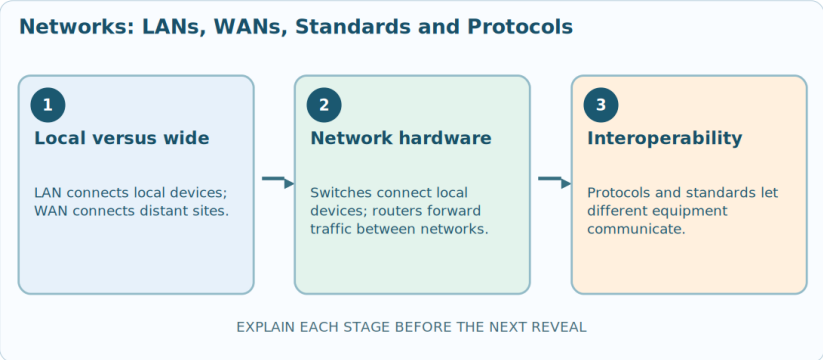
Core idea 2

Interconnect devices — Switches connect local devices; routers forward traffic between networks.

Core idea 3

Use open rules — Protocols and standards let different equipment communicate.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

LAN — Network spanning a relatively small geographical area.

WAN — Network connecting networks over larger distances.

protocol — Agreed rules for communicating between systems.

standard — Shared specification that enables interoperability.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Two classrooms share printers in the same school. Is this LAN or WAN, and why are protocols necessary?

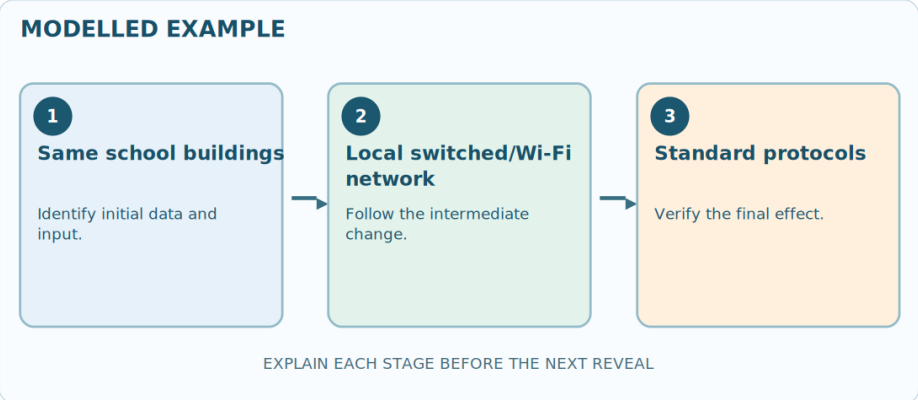
Follow the worked method

1. Same school buildings — Identify the relevant information and make a prediction.

2. Local switched/Wi-Fi network — Explain the change using correct technical terms.

3. Standard protocols — A school LAN connects local devices. Ethernet/Wi-Fi standards support compatible links and IP-based protocols allow data exchange. A router becomes relevant for communication beyond the local IP network.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Separate school LANs
Identify the relevant information and make a prediction.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Differentiate LAN and WAN with examples. [6]

Question 2

Explain why protocols and standards matter when devices use different manufacturers. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A school requires a shared printer, wireless access and broadband internet. Explain suitable network devices and their functions. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A WAN is defined by having more devices than a LAN.

Use key vocabulary from memory

1. Explain LAN in your own words.

2. Explain WAN in your own words.

3. Explain protocol in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.